

Competing Risks and Multi-State Models

Concepts, methods and software

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Part IV

Interpretation; Summary



Contents:

1. Stacked data set: advantages
2. From hazard regression to cumulative incidence
3. Which quantity to use?

Outline

Combined analysis

- Stacked data set
- Analyses
- Standard error

Advantages combined analysis

- Test for equality
- Assume equality
- Proportional baseline hazards

From regression on hazard to cumulative scale

- Competing risks

Confusion; which approach to choose?

- Choices. . .
- Etiology versus prediction
- Etiology: marginal or competing risks analysis
- Competing risks: type of quantity

Observed data

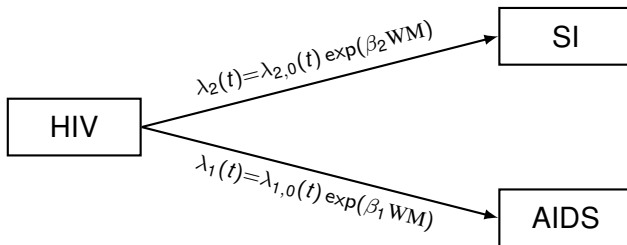
$$\{(v_1, x_1, e_1 \delta_1), \dots, (v_N, x_N, e_N \delta_N)\}$$

Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \dots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number “in risk set” (for subdistribution hazard)

Regression: covariables $\mathbf{Z}_i = (Z_{i1}, \dots, Z_{iq})$

Example: effect CCR5 Δ 32 on AIDS and SI



Analysis per event type

We cannot

- test whether effect of covariable differs per event type
- assume equality of effect for some (or all) event types
- test for proportional baseline hazards or assume them proportional

Similar issues as with stratified regression analysis

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- Competing risks: type of quantity

Stacked data set for combined analysis on cause-specific hazard

- Combine data sets for each event type

patnr	time	status	status2	cause	failcode	ccr5
14	5.05	0	0	event-free	AIDS	WW
3	2.23	1	1	AIDS	AIDS	WW
15	10.20	1	1	AIDS	AIDS	WM
8	8.61	2	0	SI	AIDS	WW
14	5.05	0	0	event-free	SI	WW
3	2.23	1	0	AIDS	SI	WW
15	10.20	1	0	AIDS	SI	WM
8	8.61	2	1	SI	SI	WW

- Baseline hazard differs by event type: stratified Cox model, with `failcode` as stratum

Stratified Cox model

Used if baseline hazard may differ per subgroup
(nonproportional hazards)

$$h_g(t | \mathbf{Z}_i) = h_{g,0}(t) \exp(\beta^\top \mathbf{Z}_i) .$$

with g subgroup (“stratum”) to which person i belongs.

Stratified Cox model

Used if baseline hazard may differ per subgroup
(nonproportional hazards)

$$h_g(t | \mathbf{Z}_i) = h_{g,0}(t) \exp(\beta^\top \mathbf{Z}_i) .$$

with g subgroup (“stratum”) to which person i belongs.
Maximise product of partial likelihoods per stratum

$$L(\beta) = \prod_{g \in \mathcal{G}} L^g(\beta)$$

with $L^g(\beta)$ partial likelihood over individuals in stratum g

- If all parameters differ by stratum, same as fitting separate Cox model per stratum
- In R:

```
coxph(Surv(time, status) ~ strata(var) + covar,
      data=...)
```

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Etiology: marginal or competing risks analysis

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Combined analysis I: interaction term

```
coxph(Surv(time, status2) ~ ccr5 * strata(failcode),
      data = aidssi.stack)
```

which yields

	coef	exp(coef)	se(coef)	z	p
ccr5WM	-1.236	0.291	0.307	-4.02	5.7e-05
ccr5WM:strata(fa	0.982	2.669	0.389	2.53	1.2e-02

Likelihood ratio test=23.2 on 2 df, p=9.29e-06 n= 648,
 number of events= 220
 (10 observations deleted due to missingness)

Compare with -1.236 and -0.254 from analysis per event type.

Analysis (in R)

```
coxph(Surv(time, status==1) ~ ccr5, data = aidssi)
```

```
-----
              coef exp(coef) se(coef)      z      p
ccr5WM.AIDS  -1.24      0.291    0.307  -4.02 5.7e-05
```

```
Likelihood ratio test=22  on 1 df, p=2.76e-06  n= 324,
                                number of events= 113
(5 observations deleted due to missingness)
```

```
coxph(Surv(time, status==2) ~ ccr5, data = aidssi)
```

```
-----
              coef exp(coef) se(coef)      z      p
ccr5WM.SI  -0.254      0.776    0.238  -1.07    0.29
```

```
Likelihood ratio test=1.19  on 1 df, p=0.275  n= 324,
                                number of events= 107
(5 observations deleted due to missingness)
```

Stacked data set for combined analysis on cause-specific hazard

- Combine data sets for each event type
and adapt covariables:

patnr	time	status	status2	cause	failcode	ccr5	ccr.comb
14	5.05	0	0	event-free	AIDS	WW	WW
3	2.23	1	1	AIDS	AIDS	WW	WW
15	10.20	1	1	AIDS	AIDS	WM	WM.AIDS
8	8.61	2	0	SI	AIDS	WW	WW
14	5.05	0	0	event-free	SI	WW	WW
3	2.23	1	0	AIDS	SI	WW	WW
15	10.20	1	0	AIDS	SI	WM	WM.SI
8	8.61	2	1	SI	SI	WW	WW

- Baseline hazard differs by event type: stratified Cox model, with `failcode` as stratum

Combined analysis II: compound variable

```
coxph(Surv(time, status2) ~ ccr.comb + strata(failcode),
      data = aidssi.stack)
```

Same effect estimates and p-values as in separate analyses

	coef	exp(coef)	se(coef)	z	p
ccr.combWM.AIDS	-1.236	0.291	0.307	-4.02	5.7e-05
ccr.combWM.SI	-0.254	0.776	0.238	-1.07	2.9e-01

Likelihood ratio test=23.2 on 2 df, p=9.29e-06 n= 648,
 number of events= 220
 (10 observations deleted due to missingness)

Analysis (in R)

```
coxph(Surv(time, status==1) ~ ccr5, data = aidssi)
```

```
-----
              coef exp(coef) se(coef)      z      p
ccr5WM.AIDS  -1.24      0.291    0.307  -4.02 5.7e-05
```

```
Likelihood ratio test=22  on 1 df, p=2.76e-06  n= 324,
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coxph(Surv(time, status==2) ~ ccr5, data = aidssi)
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              coef exp(coef) se(coef)      z      p
ccr5WM.SI  -0.254      0.776    0.238  -1.07  0.29
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Likelihood ratio test=1.19  on 1 df, p=0.275  n= 324,
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Stacked data set for combined analysis on cause-specific hazard

- Combine data sets for each event type
and adapt covariables:

patnr	time	status	status2	cause	failcode	ccr5	ccr.comb	ccrAIDS	ccrSI
14	5.05	0	0	event-free	AIDS	WW	WW	0	0
3	2.23	1	1	AIDS	AIDS	WW	WW	0	0
15	10.20	1	1	AIDS	AIDS	WM	WM.AIDS	1	0
8	8.61	2	0	SI	AIDS	WW	WW	0	0
14	5.05	0	0	event-free	SI	WW	WW	0	0
3	2.23	1	0	AIDS	SI	WW	WW	0	0
15	10.20	1	0	AIDS	SI	WM	WM.SI	0	1
8	8.61	2	1	SI	SI	WW	WW	0	0

- Baseline hazard differs by event type: stratified Cox model, with `failcode` as stratum

Combined analysis III: type-specific covariables

- The variables `ccrSI` and `ccrAIDS` are examples of **type-specific** covariables
- Instead of specifying (in separate models k)

$$\lambda_k(t \mid \mathbf{Z}_i) = \lambda_{k,0}(t) \exp(\boldsymbol{\beta}_k^\top \mathbf{Z}_i),$$

we can equivalently specify (in one model)

$$\lambda_k(t \mid \mathbf{Z}_i) = \lambda_{k,0}(t) \exp(\boldsymbol{\beta}^\top \mathbf{Z}_{k,i}),$$

- $\mathbf{Z}_{k,i}$ equals $\mathbf{Z}_i$ for cause k , and has value 0 otherwise

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$$\lambda_k(t | \mathbf{Z}_i) = \lambda_{k,0}(t) \exp(\beta_k^\top \mathbf{Z}_i),$$

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$$\lambda_k(t | \mathbf{Z}_i) = \lambda_{k,0}(t) \exp(\beta^\top \mathbf{Z}_{k,i}),$$

- $\mathbf{Z}_{k,i}$ equals $\mathbf{Z}_i$ for cause k , and has value 0 otherwise
- Eg. $Z_i = \text{CCR5 genotype coding (WM=1, WW=0)}$;
AIDS: $k = 1$; SI: $k = 2$:

$$\text{ccrAIDS}_i = Z_{1,i} = \begin{cases} Z_i, & \text{if failcode=AIDS} \\ 0, & \text{if failcode=SI} \end{cases}$$

and

$$\text{ccrSI}_i = Z_{2,i} = \begin{cases} 0, & \text{if failcode=AIDS} \\ Z_i, & \text{if failcode=SI} \end{cases}$$

Combined analysis III: results

```
coxph(Surv(time, status2) ~ ccrAIDS + ccrSI +
      strata(failcode), data = aidssi.stack)
```

Same effect estimates and p-values as before

	coef	exp(coef)	se(coef)	z	p
ccrAIDS	-1.236	0.291	0.307	-4.02	5.7e-05
ccrSI	-0.254	0.776	0.238	-1.07	2.9e-01

Why does this work?

Partial likelihood for $\beta = (\beta_1, \beta_2) = (\beta_{\text{AIDS}}, \beta_{\text{SI}})$ with `failcode` as stratum: $L(\beta) = L^{\text{AIDS}}(\beta) \times L^{\text{SI}}(\beta)$

$(R_{\text{AIDS}}(t_i)/R_{\text{SI}}(t_i))$: risk set under `failcode=AIDS/SI`

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$(R_{\text{AIDS}}(t_i)/R_{\text{SI}}(t_i))$: risk set under `failcode=AIDS/SI`

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1 \dots 324 \\ \text{status2}_i=1}} \left\{ \frac{\exp(\beta_1 \times \text{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta_1 \times \text{ccrAIDS}_j + \beta_2 \times 0)} \right\} \text{aidssi.stack}$$

Why does this work?

Partial likelihood for $\beta = (\beta_1, \beta_2) = (\beta_{\text{AIDS}}, \beta_{\text{SI}})$ with failcode as stratum: $L(\beta) = L^{\text{AIDS}}(\beta) \times L^{\text{SI}}(\beta)$

$(R_{\text{AIDS}}(t_i)/R_{\text{SI}}(t_i))$: risk set under failcode=AIDS/SI

$$\begin{aligned}
 L^{\text{AIDS}}(\beta) &= \prod_{\substack{i=1 \dots 324 \\ \text{status}_{2i}=1}} \left\{ \frac{\exp(\beta_1 \times \text{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta_1 \times \text{ccrAIDS}_j + \beta_2 \times 0)} \right\} \text{aidssi.stack} \\
 &= \prod_{\substack{i=1 \dots 324 \\ \text{status}_i=1}} \left\{ \frac{\exp(\beta_1 \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_1 \times \text{ccr5}_j)} \right\} \text{aidssi}
 \end{aligned}$$

Why does this work?

Partial likelihood for $\beta = (\beta_1, \beta_2) = (\beta_{\text{AIDS}}, \beta_{\text{SI}})$ with failcode as stratum: $L(\beta) = L^{\text{AIDS}}(\beta) \times L^{\text{SI}}(\beta)$

$(R_{\text{AIDS}}(t_i)/R_{\text{SI}}(t_i))$: risk set under failcode=AIDS/SI

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1 \dots 324 \\ \text{status}_{2i}=1}} \left\{ \frac{\exp(\beta_1 \times \text{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta_1 \times \text{ccrAIDS}_j + \beta_2 \times 0)} \right\} \text{ aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \text{status}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_1 \times \text{ccr5}_j)} \right\} \text{ aidssi}$$

$$L^{\text{SI}}(\beta) = \prod_{\substack{i=325 \dots 648 \\ \text{status}_{2i}=1}} \left\{ \frac{\exp(\beta_1 \times 0 + \beta_2 \times \text{ccrSI}_i)}{\sum_{j \in R_{\text{SI}}(t_i)} \exp(\beta_1 \times 0 + \beta_2 \times \text{ccrSI}_j)} \right\} \text{ aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \text{status}_i=2}}^{324} \left\{ \frac{\exp(\beta_2 \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_2 \times \text{ccr5}_j)} \right\} \text{ aidssi}$$

Why does this work?

Partial likelihood for $\beta = (\beta_1, \beta_2) = (\beta_{\text{AIDS}}, \beta_{\text{SI}})$ with failcode as stratum: $L(\beta) = L^{\text{AIDS}}(\beta) \times L^{\text{SI}}(\beta)$

$(R_{\text{AIDS}}(t_i)/R_{\text{SI}}(t_i))$: risk set under failcode=AIDS/SI

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1 \dots 324 \\ \text{status}_{2i}=1}} \left\{ \frac{\exp(\beta_1 \times \text{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta_1 \times \text{ccrAIDS}_j + \beta_2 \times 0)} \right\} \quad \text{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \text{status}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_1 \times \text{ccr5}_j)} \right\} \quad \text{aidssi}$$

$$L^{\text{SI}}(\beta) = \prod_{\substack{i=325 \dots 648 \\ \text{status}_{2i}=1}} \left\{ \frac{\exp(\beta_1 \times 0 + \beta_2 \times \text{ccrSI}_i)}{\sum_{j \in R_{\text{SI}}(t_i)} \exp(\beta_1 \times 0 + \beta_2 \times \text{ccrSI}_j)} \right\} \quad \text{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \text{status}_{i=2}}}^{324} \left\{ \frac{\exp(\beta_2 \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_2 \times \text{ccr5}_j)} \right\} \quad \text{aidssi}$$

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Combined analysis

Stacked data set

Analyses

Standard error

Advantages combined analysis

Test for equality

Assume equality

Proportional baseline hazards

From regression on hazard to cumulative scale

Competing risks

Confusion; which approach to choose?

Choices...

Etiology versus prediction

Etiology: marginal or competing risks analysis

Competing risks: type of quantity

Robust estimate of standard error?

```
coxph(Surv(time, status) ~ ccr.comb + strata(failcode) +
      cluster(patnr), data = aidssi.stack)
```

gives

	coef	se(coef)	robust se	z	p
ccr.combWM.AIDS	-1.236	0.307	0.296	-4.18	0.00003
ccr.combWM.SI	-0.254	0.238	0.233	-1.09	0.27000

Robust estimate differs from separate analyses

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Advantage I: test whether effects are equal

```
coxph(Surv(time, status) ~ ccr5 + ccrSI + strata(failcode),
      aidssi.stack)
```

gives

	coef	exp(coef)	se(coef)	z	p
ccr5WM	-1.236	0.291	0.307	-4.02	5.7e-05
ccrSI	0.982	2.669	0.389	2.53	1.2e-02

Note: same result via interaction term:

```
coxph(Surv(time, status) ~ ccr5 * strata(failcode),
      data = aidssi.stack)
```

Explanation model

patnr	time	status	failcode	ccr5	ccr.comb	ccrAIDS	ccrSI
14	5.05	0	AIDS	WW	WW	0	0
3	2.23	1	AIDS	WW	WW	0	0
15	10.20	1	AIDS	WM	WM.AIDS	1	0
8	8.61	0	AIDS	WW	WW	0	0
14	5.05	0	SI	WW	WW	0	0
3	2.23	0	SI	WW	WW	0	0
15	10.20	0	SI	WM	WM.SI	0	1
8	8.61	1	SI	WW	WW	0	0

$$\lambda_k(t) = \lambda_{k,0}(t) \exp(\beta_1 \times \text{ccr5} + \beta_2 \times \text{ccrSI}).$$

AIDS; WW

$$\lambda_{1,0}(t)$$

AIDS; WM

$$\lambda_{1,0}(t) \exp(\beta_1)$$

SI; WW

$$\lambda_{2,0}(t)$$

SI; WM

$$\lambda_{2,0}(t) \exp(\beta_1 + \beta_2)$$

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Advantage II: assume equality

Same effect of CCR5 mutation on relative hazards for both transitions

```
coxph(Surv(time, status) ~ ccr5 + strata(failcode),
      data = aidssi.stack)
```

	coef	exp(coef)	se(coef)	z	p
ccr5	-0.701	0.496	0.186	-3.77	0.00016

Advantage II: assume equality

Same effect of CCR5 mutation on relative hazards for both transitions

```
coxph(Surv(time, status) ~ ccr5 + strata(failcode),
      data = aidssi.stack)
```

	coef	exp(coef)	se(coef)	z	p
ccr5	-0.701	0.496	0.186	-3.77	0.00016

This is similar to

```
coxph(Surv(time, status!=0) ~ ccr5, data = aidssi)
```

with original data set

patnr	time	status	ccr5
14	5.05	0	WW
3	2.23	1	WW
15	10.20	1	WM
8	8.61	2	WW

Why?

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1 \dots 324 \\ \text{status}_{2i}=1}} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta \times \text{ccr5}_j)} \right) \quad \text{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \text{status}_i=1}}^{324} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \text{ccr5}_j)} \right) \quad \text{aidssi}$$

$$L^{\text{SI}}(\beta) = \prod_{\substack{i=325 \dots 648 \\ \text{status}_{2i}=1}} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R_{\text{SI}}(t_i)} \exp(\beta \times \text{ccr5}_j)} \right) \quad \text{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \text{status}_i=2}}^{324} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \text{ccr5}_j)} \right) \quad \text{aidssi}$$

Why?

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1 \dots 324 \\ \text{status}_i=1}} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta \times \text{ccr5}_j)} \right) \quad \text{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \text{status}_i=1}}^{324} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \text{ccr5}_j)} \right) \quad \text{aidssi}$$

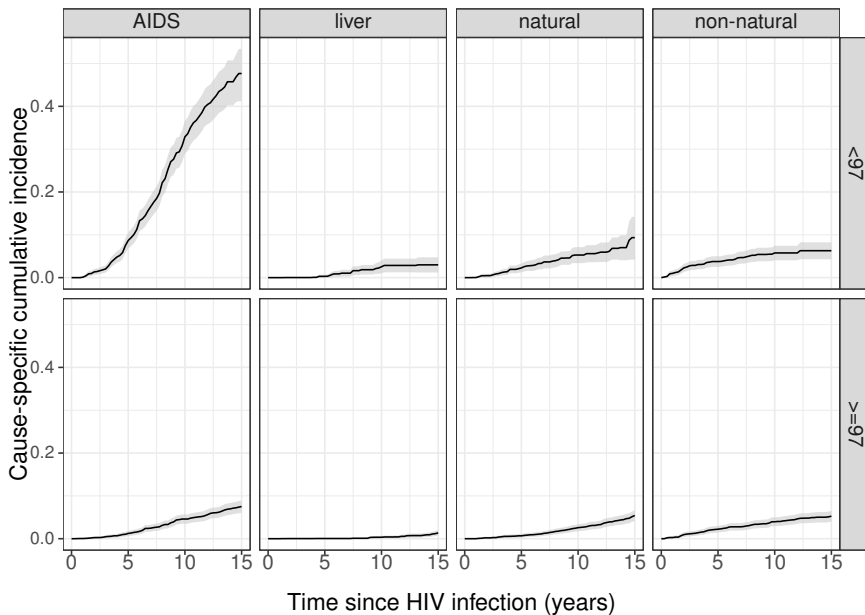
$$L^{\text{SI}}(\beta) = \prod_{\substack{i=325 \dots 648 \\ \text{status}_i=1}} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R_{\text{SI}}(t_i)} \exp(\beta \times \text{ccr5}_j)} \right) \quad \text{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \text{status}_i=2}}^{324} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \text{ccr5}_j)} \right) \quad \text{aidssi}$$

Same as partial likelihood based on aidssi data:

$$\prod_{\substack{i=1 \\ \text{status}_i>0}}^{324} \left(\frac{\exp(\beta \times \text{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \text{ccr5}_j)} \right)$$

Cause-specific mortality by calendar period



Assuming equality necessary when some event types are rare

$N = 9164$, correct for age, sex, risk group, calendar period

	MSM	idu	haemophilic	MSW	total
alive	5052	1310	73	2252	8446
aids	158	148	65	24	397
liver	3	15	22	1	39
non natural	24	98	2	6	128
natural	68	47	14	23	154

Too few events for proper analysis per event type

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Advantage III: Proportional baseline hazards

$$\lambda_{\text{SI},0}(t) = \lambda_{\text{AIDS},0}(t) \times \exp(\alpha)$$

Advantage III: Proportional baseline hazards

$$\lambda_{SI,0}(t) = \lambda_{AIDS,0}(t) \times \exp(\alpha)$$

Total model now reads (AIDS: $k = 1$; SI: $k = 2$):

$$\lambda_k(t | \mathbf{Z}_i) = \lambda_{1,0}(t) \exp\{\alpha \times (k-1) + \beta_1 \times \text{ccrAIDS}_i + \beta_2 \times \text{ccrSI}_i\} \quad (1)$$

Advantage III: Proportional baseline hazards

$$\lambda_{SI,0}(t) = \lambda_{AIDS,0}(t) \times \exp(\alpha)$$

Total model now reads (AIDS: $k = 1$; SI: $k = 2$):

$$\lambda_k(t | \mathbf{Z}_i) = \lambda_{1,0}(t) \exp\{\alpha \times (k-1) + \beta_1 \times \text{ccrAIDS}_i + \beta_2 \times \text{ccrSI}_i\} \quad (1)$$

In R:

```
coxph(Surv(time, status) ~ ccrAIDS + ccrSI +  
      failcode, data = aidssi.stack)
```

Results:

	coef	exp(coef)	se(coef)	z	p
ccrAIDS	-1.166	0.311	0.306	-3.81	0.00014
ccrSI	-0.332	0.718	0.237	-1.40	0.16000
failcodeSI	-0.184	0.832	0.148	-1.25	0.21000

```
Likelihood ratio test=21.5 on 3 df, p=8.12e-05 n= 648,  
number of events= 220  
(10 observations deleted due to missingness)
```

Outline

Combined analysis

- Stacked data set
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From regression on hazard to cumulative scale

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Confusion; which approach to choose?

- Choices...
- Etiology versus prediction
- Etiology: marginal or competing risks analysis
- Competing risks: type of quantity

Observed data

$$\{(v_1, x_1, e_1 \delta_1), \dots, (v_N, x_N, e_N \delta_N)\}$$

Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \dots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number “in risk set” (for subdistribution hazard)

Regression: covariables $\mathbf{Z}_i = (Z_{i1}, \dots, Z_{iq})$

Standard survival setting: one event type

- Cox regression, $\mathbf{Z}$ time-fixed: $h(t) = h_0(t) \exp(\beta^\top \mathbf{Z}_i)$
- One-to-one relation between effects on hazard and survival

$$\bar{F}(t|\mathbf{Z}_i) = \exp\left\{-\int_0^t h_0(s) \exp(\beta^\top \mathbf{Z}_i) ds\right\} = \bar{F}_0(t)^{\exp(\beta^\top \mathbf{Z}_i)},$$

with $\bar{F}_0(t) = \exp\left\{-\int_0^t h_0(s) ds\right\} = \exp\{-H_0(t)\}$

- Estimate becomes

$$\hat{\bar{F}}(t|\mathbf{Z}_i) = \exp\left\{-\hat{H}_0(t) e^{\hat{\beta}^\top \mathbf{Z}_i}\right\} = \hat{\bar{F}}_0(t)^{\exp(\hat{\beta}^\top \mathbf{Z}_i)}$$

with $\hat{\bar{F}}_0(t) = \exp\{-\hat{H}_0(t)\}$ estimated “baseline” survival curve

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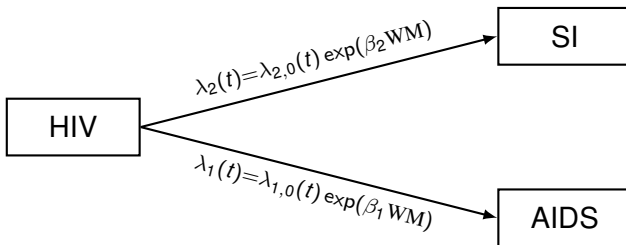
From regression on hazard to cumulative scale

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Example: effect CCR5 Δ 32 on AIDS and SI



Cause-specific hazard

- Cumulative progression to one end point also depends on progression to other end points
- “No one-to-one relation with cumulative scale”
- Example C: CCR5-Δ32 deletion and SI switch

$$F_{SI}(t) = P\{T \leq t, E = SI\} = \int_0^t P\{T \geq s\} \lambda_{SI}(s) ds$$

$$\text{with } P\{T \geq s\} = \exp\left[-\int_0^s \{\lambda_{SI}(u) + \lambda_{AIDS}(u)\} du\right]$$

Estimate based on proportional cause-specific hazards

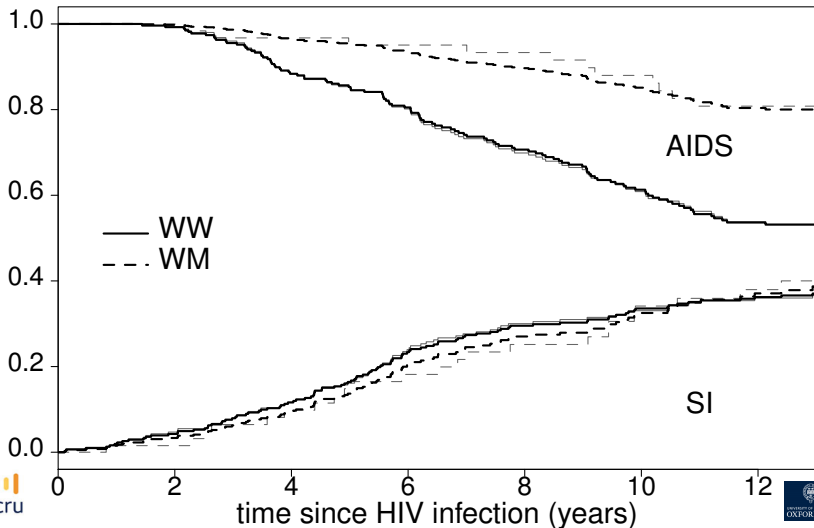
$$\begin{aligned}\widehat{F}_{\text{SI}}(t \mid \text{ccr5}) &= \sum_{t_{(i)} \leq t} \widehat{\lambda}_{\text{SI}}(t_{(i)} \mid \text{ccr5}) \times \widehat{\overline{F}}(t_{(i)} - \mid \text{ccr5}) \\ &= \sum_{t_{(i)} \leq t} \widehat{\lambda}_{0,\text{SI}}(t_{(i)}) \exp(\widehat{\beta}_{\text{SI}} \text{ccr5}) \times \\ &\quad \exp\left\{-\left[\widehat{\Lambda}_{\text{AIDS}}(t_{(i)} - \mid \text{ccr5}) + \widehat{\Lambda}_{\text{SI}}(t_{(i)} - \mid \text{ccr5})\right]\right\}\end{aligned}$$

and use

$$\begin{aligned}\widehat{\Lambda}_{\text{SI}}(t \mid \text{ccr5}) &= \sum_{t_{(i)} \leq t} \widehat{\lambda}_{\text{SI}}(t_{(i)} \mid \text{ccr5}) \\ &= \sum_{t_{(i)} \leq t} \widehat{\lambda}_{0,\text{SI}}(t_{(i)}) \exp(\widehat{\beta}_{\text{SI}} \text{ccr5}) \\ &= \widehat{\Lambda}_{0,\text{SI}}(t) \exp(\widehat{\beta}_{\text{SI}} \text{ccr5}).\end{aligned}$$

and similarly $\widehat{\Lambda}_{\text{AIDS}}(t \mid \text{ccr5}) = \widehat{\Lambda}_{0,\text{AIDS}}(t) \exp(\widehat{\beta}_{\text{AIDS}} \text{ccr5})$

Cumulative incidence based on proportional cause-specific hazards model



Standard survival setting: one event type

- Cox regression, $\mathbf{Z}$ time-fixed: $h(t) = h_0(t) \exp(\beta^\top \mathbf{Z}_i)$
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$$\bar{F}(t|\mathbf{Z}_i) = \exp\left\{-\int_0^t h_0(s) \exp(\beta^\top \mathbf{Z}_i) ds\right\} = \bar{F}_0(t)^{\exp(\beta^\top \mathbf{Z}_i)},$$

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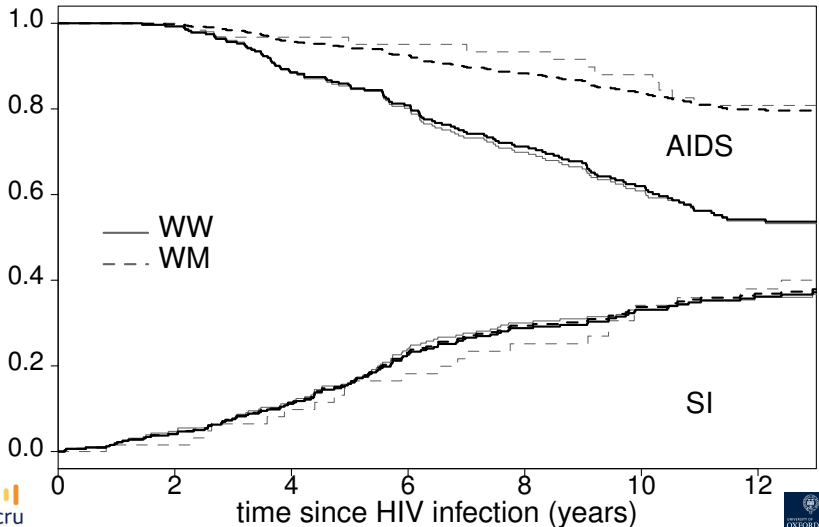
- Estimate becomes

$$\hat{\bar{F}}(t|\mathbf{Z}_i) = \exp\left\{-\hat{H}_0(t) e^{\hat{\beta}^\top \mathbf{Z}_i}\right\} = \hat{\bar{F}}_0(t)^{\exp(\hat{\beta}^\top \mathbf{Z}_i)}$$

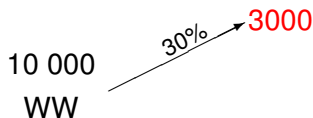
with $\hat{\bar{F}}_0(t) = \exp\{-\hat{H}_0(t)\}$ estimated “baseline” survival curve

- Similar for regression model on subdistribution hazard

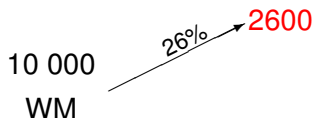
Cumulative incidence based on proportional subdistribution hazards model



“Reproducing” this result: CS-RH $\neq$ cumulative effect



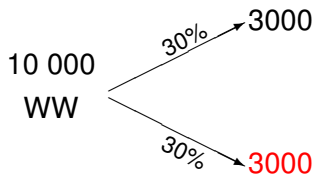
SI



SI

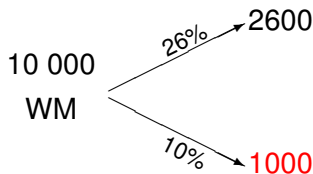
$$\text{CS} - \text{RH}(\text{SI}) = 26/30 = 0.87$$

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SI

AIDS



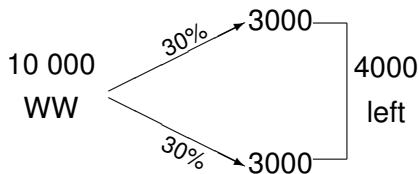
SI

AIDS

$$\text{CS} - \text{RH}(\text{AIDS}) = 10/30 = 0.33$$

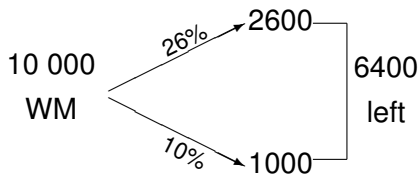
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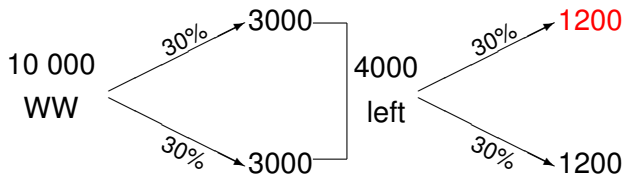
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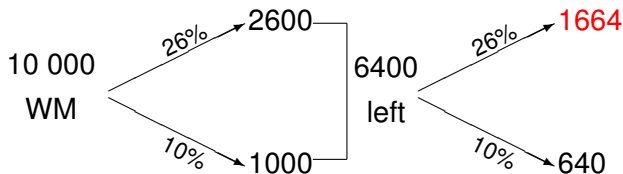
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AIDS



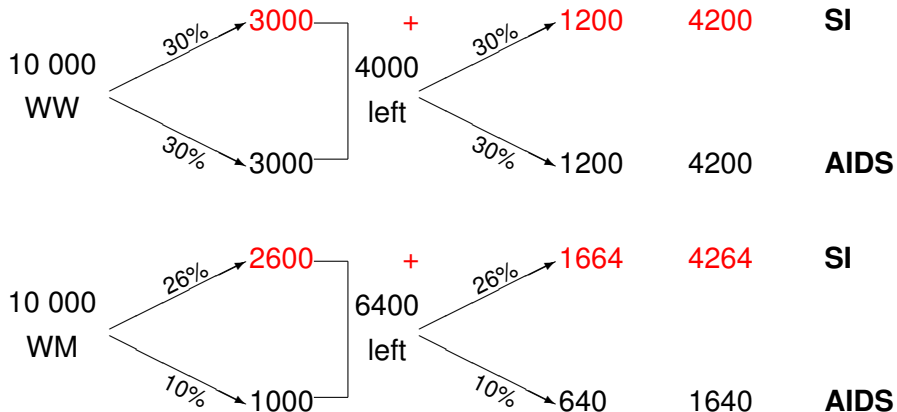
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“Reproducing” this result: CS-RH $\neq$ cumulative effect



$$\text{CS} - \text{RH}(\text{AIDS}) = 10/30 = 0.33$$

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Choices...

- Marginal analysis or competing risks analysis?
- Hazard or cumulative probability?
- Competing risks: cause-specific hazard or subdistribution hazard?

Example: complications of radiation therapy. Perspective of patient or oncologist?

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Purpose of study

- Prediction
 - Cumulative event probability most relevant
 - Usually refers to real world, in which competing risks exist
- Etiology: causal relations between exposure and outcome
 - Causality (see Gillies, 2019)
 - Actions, interventions and manipulations
 - Biological mechanisms
 - May refer to counterfactual world without competing risks

Choices...

- Marginal analysis or competing risks analysis?
- Hazard or cumulative probability?
- Competing risks: cause-specific hazard or subdistribution hazard?

Example: complications of radiation therapy. Perspective of patient or oncologist?

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Intervention or biological mechanism?

- Competing risks an intervention: marginal setting can be imagined
- Competing risk biological but unrelated mechanism: marginal
- Competing risk part of same mechanism: competing risks

Exercise 23.1.2

A competing risk is an event whose occurrence either precludes the occurrence of the event under examination or fundamentally alters the probability of occurrence of that event.

Choices...

- Marginal analysis or competing risks analysis?
- Hazard or cumulative probability?
- Competing risks: cause-specific hazard or subdistribution hazard?

Example: complications of radiation therapy. Perspective of patient or oncologist?

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Cause-specific or subdistribution hazard?

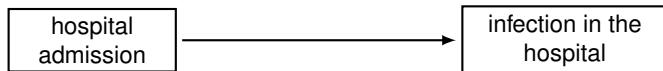
- Both describe competing risks setting
- Cause-specific often closer to causal effects, but any hazard is problematic
- Subdistribution directly relates to cumulative scale
 - Combination of etiology and effect on other event types. “Impact” instead of “effect”
 - Regression: impact on cumulative scale via parameters
- Some suggest to quantify both

Hazard or cumulative probability?

- Prediction: cumulative probability, but may be based on regression model for the hazard
- Etiology: hazards have been shown to be problematic. Lots of recent developments.
- Cumulative probability: regression
 - Via cause-specific hazards, but no one-to-one relation
 - Via subdistribution hazards, one-to-one relation with cumulative scale
 - Directly via cause-specific cumulative incidence (not covered in this course)

Which to choose depends on data characteristics and ease of presentation

G: Staphylococcus infection during hospital stay



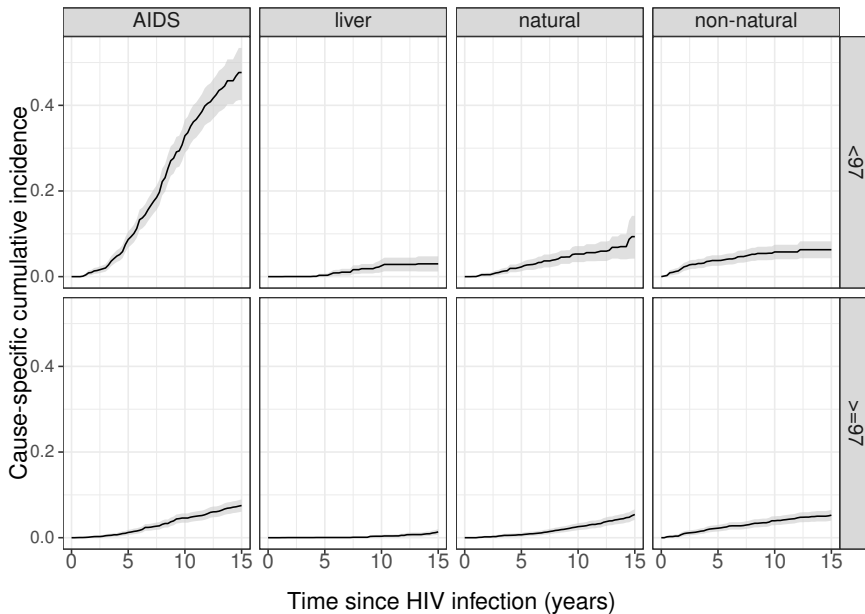
- Etiology (biological question): infection risk in hospital. What would happen if everyone stayed in hospital (no discharge)?
 - marginal distribution/net risk
- Predict (clinical question): burden due to infection while in hospital; discharge prevents event to occur
 - cause-specific cumulative incidence /subdistribution/crude risk

- Which of two hospitals has higher risk may depend on type of risk

The examples revisited

- Example 1: staphylococcus infection in hospital
 - Etiology (what if everyone would stay in hospital): marginal hazard, Kaplan-Meier.
Assumption: competing risk discharge independent
 - Prediction (how many infections are observed in hospital): cause-specific cumulative incidence

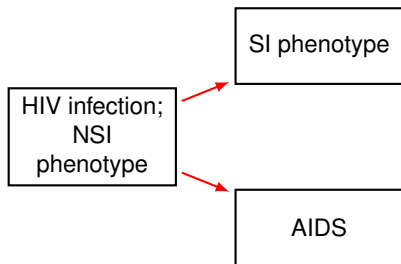
Cause-specific mortality by calendar period



The examples revisited

- Example 1: staphylococcus infection in hospital
 - Etiology (what if everyone would stay in hospital): marginal hazard, Kaplan-Meier.
Assumption: competing risk discharge independent
 - Prediction (how many infections are observed in hospital): cause-specific cumulative incidence
- Example 2: impact of cART on causes of death.
 - Competing risks
 - Etiology: cause-specific hazard(?)
Prediction: cause-specific cumulative incidence;
subdistribution hazard for regression model

C: AIDS and SI switch after HIV infection



- SI: “syncytium inducing phenotype”. At HIV infection only NSI
- Competing risks: time to either SI switch or AIDS **as first event**

The examples revisited

- Example 1: staphylococcus infection in hospital
 - Etiology (what if everyone would stay in hospital): marginal hazard, Kaplan-Meier.
Assumption: competing risk discharge independent
 - Prediction (how many infections are observed in hospital): cause-specific cumulative incidence
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 - Etiology: cause-specific hazard(?)
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- Example 3: SI and AIDS as first event after HIV infection.
 - Competing risks. Marginal analysis possible if other event is ignored
 - Etiology: cause-specific hazard(?)
Prediction: cause-specific cumulative incidence

The examples revisited

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Assumption: competing risk discharge independent
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- Example 3: SI and AIDS as first event after HIV infection.
 - Competing risks. Marginal analysis possible if other event is ignored
 - Etiology: cause-specific hazard(?)
Prediction: cause-specific cumulative incidence
- Example 4: natural history in IDU en MSM.
 - Marginal distribution
 - Problem: competing risks are dependent

Exercise 23.1.5

Study on the effect of the use of β -blockers on prostate cancer (PCa). Individuals that used β -blockers had lower subdistribution hazard for PCa-specific death but higher for DOC.

To address this potential bias, we performed all analyses with the Fine and Gray competing risk regression model. In addition, we observed no increase in all-cause mortality among β -blocker users although other-cause mortality was higher, strengthening the interpretation of an association between the use of β -blockers and PCa-specific mortality.

Exercise 23.1.5

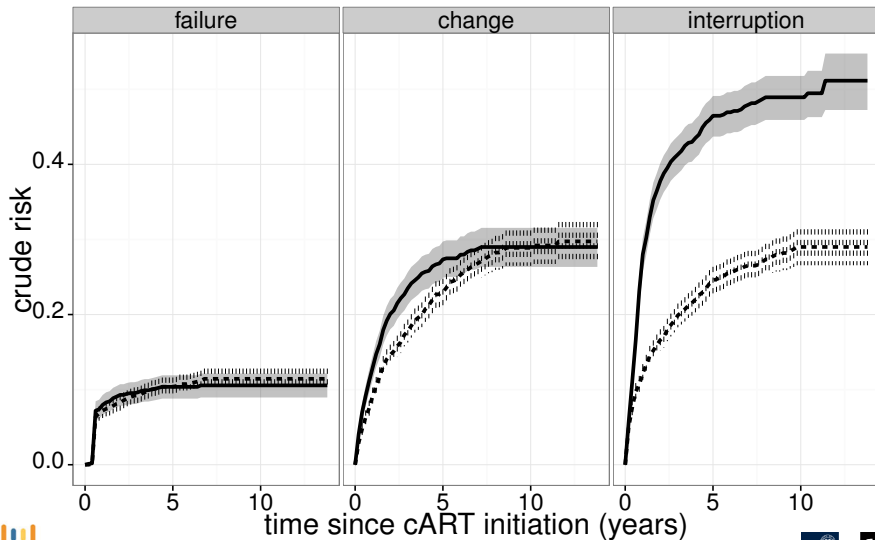
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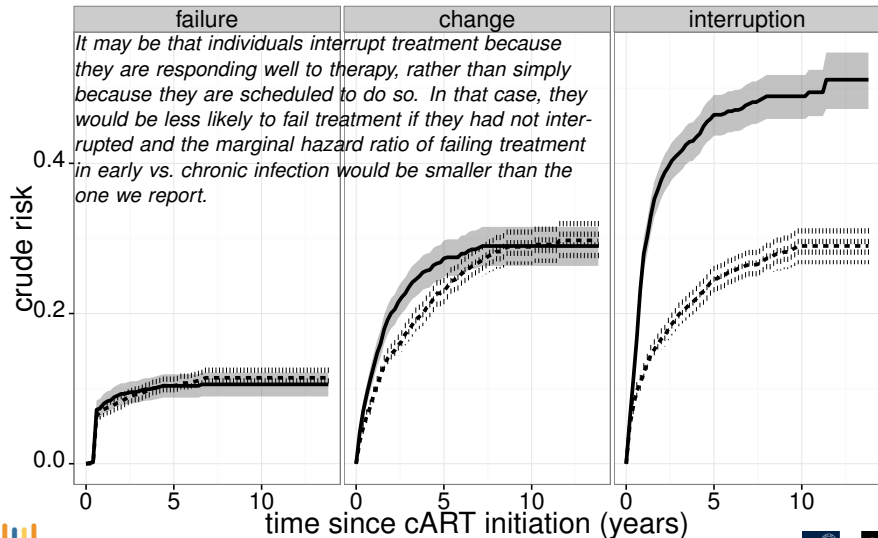
They want to study etiology, which is better described by cause-specific hazards.

H.H. Grytli *et al.*. Association between use of β -blockers and prostate cancer-specific

Exercise 23.1.6: cART; early versus late starters



Exercise 23.1.6: cART; early versus late starters



Answer

- Assume that treatment failure and treatment interruption are the only two competing events
- Most extreme scenario: those who interrupt treatment will never fail. Marginal hazard same as the subdistribution hazard, i.e. the reported one.
- It may be true if the effect was observed for the cause-specific hazard.